In [1]:
```
# Install once if needed:
# pip install gensim

# 1. Import libraries
import re
from gensim.models import Word2Vec
```

In [2]:
```
# 2. Enter the given textual document
text = """
The speed of transmission is an important point of difference between the two viruses.
Influenza has a shorter median incubation period and a shorter serial interval than COVID-19.
The serial interval for COVID-19 virus is estimated to be 5-6 days, while for influenza virus,
the serial interval is 3 days. This means that influenza can spread faster than COVID-19.

Further, transmission in the first 3-5 days of illness, or potentially pre-symptomatic
transmission, is a major driver of transmission for influenza. In contrast, while there are
people who can shed COVID-19 virus 24-48 hours prior to symptom onset, this does not appear
to be a major driver of transmission.

The reproductive number is understood to be between 2 and 2.5 for COVID-19 virus,
higher than for influenza. However, estimates for both COVID-19 and influenza viruses
are very context and time-specific, making direct comparisons more difficult.
"""
```

In [3]:
```
# 3. Clean and tokenize the document
def clean_text(text):
    text = text.lower()
    text = re.sub(r"[^a-z\s]", " ", text)
    return text.split()

sentences = [clean_text(text)]
```

In [4]:
```
# 4. Train CBOW Word2Vec model (sg=0 means CBOW)
cbow_model = Word2Vec(
    sentences=sentences,
    vector_size=50,
    window=3,
    min_count=1,
```

```
    sg=0,
    workers=1,
    epochs=100
)
```

In [5]:
```
# 5. Display vocabulary size
print("Vocabulary Size:", len(cbow_model.wv))

# 6. Display the embedding of a word
word = "influenza"

if word in cbow_model.wv:
    print("Embedding for", word, ":")
    print(cbow_model.wv[word])
```

```
Vocabulary Size: 76
Embedding for influenza :
[ 0.00726056  0.00298959  0.0209852   0.02932329 -0.01638395 -0.04717096
  0.0538315   0.08585211 -0.02404951 -0.0650765  -0.00185414 -0.09499653
  0.04091202  0.0733184  -0.0320023   0.0631863   0.06264899  0.05317801
 -0.06117572 -0.06477513  0.04613476  0.0360618   0.0881709  -0.0241031
  0.04046386  0.0283855  -0.03358545  0.04989163 -0.0260111   0.05057472
  0.00560099 -0.05216631  0.02617306 -0.06867228 -0.02300332  0.0203191
  0.07026142 -0.00290394  0.02038599  0.05565857  0.05069904 -0.04395452
 -0.02085973 -0.00470997  0.08071746 -0.00636815  0.00451803 -0.05115084
  0.04146405  0.0382991 ]
```

In [6]:
```
# 7. Find similar words
if word in cbow_model.wv:
    print("Similar words to", word, ":")
    print(cbow_model.wv.most_similar(word, topn=5))
```

```
Similar words to influenza :
[('for', 0.9524333477020264), ('covid', 0.9507610201835632), ('virus', 0.9429670572280884), ('is', 0.9364709258079529), ('of',
0.9339848160743713)]
```

In [ ]: